

# Bhaumik Bhalara

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## SUMMARY

Aspiring Electrical and Electronics Engineer with a strong foundation in Verilog, Python, and FPGA hardware design. Proven ability to contribute to innovative projects with a focus on performance optimization and efficiency. Seeking to leverage technical skills in a dynamic organization committed to continuous learning and pioneering advancements in technology.

## EXPERIENCE

### Intern

3D Cubic, Surat

Aug 2021 - Oct 2021

- Completed an intensive internship focused on understanding and operating 3D printing technology, enhancing practical knowledge in additive manufacturing processes.
- Designed and fabricated a 2 cm x 2 cm x 2 cm polyamide cube using a 3D printer, demonstrating hands-on expertise and attention to detail in prototyping and design execution.

## EDUCATION

### Bachelor of Engineering

Sarvajani College of Engineering and Technology

May 2022

GPA: 3.30/4

## SKILLS

Verilog, Python, FPGA Design, SystemVerilog, TCL, Perl, Cadence, Synopsys VCS, Primetime, Quartus II IDE, MATLAB, G-Code, PCI-E, PCI, Intel DE-10 Lite FPGA Board, Oscilloscopes, DMM, Signal Generators, Sensors, Probes, Bread Board, Simplify3D, MobaXterm

## PROJECTS

### 64-bit PCI to SRAM Memory Card

Aug 2023 - Dec 2023

- Developed SRAM architecture with detailed block and state diagrams for PCI to SRAM transactions, achieving a seamless data flow with a focus on performance and reliability.
- Prepared comprehensive timing diagrams for both 64-64 bit and 32-64-bit bus transactions, ensuring adherence to design specifications and enhancing system efficiency.

### Vending Machine

Aug 2023 - Dec 2023

- Integrated finite state machines, including idle, selection, payment, dispensing, and reset states, to ensure smooth operation and user interaction.
- Tested and verified the design using Verilog testbenches and simulation environments, successfully validating functionality on an Intel DE-10 Lite FPGA board, impacting user experience positively.

### 6-Bit Segmented DAC

Jan 2024 - May 2024

- Divided DAC implementation into two segments: LSB and MSB, enhancing signal integrity and reducing glitch noise by 30%.
- Achieved expected results for both theoretical and simulated values of 12V, ensuring high accuracy in signal output.

### STA for Microprocessor Netlist

Aug 2024 - Dec 2024

- Executed design constraints script to generate comprehensive setup and hold timing reports, identifying critical timing paths and potential violations.
- Optimized generated slack to improve overall design performance by 25%, enhancing system reliability and efficiency.

### 4-Bit Ripple Carry Adder

Aug 2024 - Dec 2024

- Implemented Verilog module and Perl script to verify the functionality of the adder, achieving 100% accuracy in test results.
- Generated timing reports by applying design constraints, utilizing switches for static timing analysis, which improved design validation processes.